

Key References

- 1. Raissi, Perdikaris & Karniadakis (2019).** Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear PDEs. *J. Comput. Phys.* **378**, 686–707.
Foundational PINN formulation — the data + physics-residual loss used in every model here.
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Closest template to this work: inverse PINNs on biological ODE networks; the local-sensitivity/Fisher route to identifiability.
- 3. Yang, Meng & Karniadakis (2021).** B-PINNs: Bayesian physics-informed neural networks for forward and inverse PDE problems with noisy data. *J. Comput. Phys.* **425**, 109913.
Basis of the Bayesian inverse PINN: HMC over parameters gives posterior marginals instead of a point estimate.
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The identifiability verdict — profile likelihood distinguishes the structural from the practical (high-WNT θ_P) wall.
- 5. Wang, Yu & Perdikaris (2022).** When and why PINNs fail to train: a neural tangent kernel perspective. *J. Comput. Phys.* **449**, 110768.
Explains the loss-term imbalance and spectral bias behind the training pathologies; motivates adaptive loss weighting.
- 6. Tancik et al. (2020).** Fourier features let networks learn high frequency functions in low dimensional domains. *NeurIPS* **33**, 7537–7547.
Fourier-feature time embedding — required to capture the circadian RA forcing and the ATRA pulse.
- 7. Ji, Qiu, Zhang & Deng (2021).** Stiff-PINN: physics-informed neural network for stiff chemical kinetics. *J. Phys. Chem. A* **125**(36), 8098–8106.
Diagnoses PINN failure on stiff multiscale kinetics — the same stiffness that shapes the WNT–RA–HOX residual.